

ModalPINN EnKF Audit, Part 2 — Final Evaluation and the Observability Answer

Branch `enkf-repairs`. Re=100 cylinder wake, 32 wall pressure taps, no velocity data anywhere. Domain $[-4,8] \times [-4,4]$, $\omega_0 = 1.036$.

Every number below was recomputed for this report from the run archives; none is copied from a previous log. The repaired nominal run was re-executed from scratch (`stage_d2_enkf_nominal_repro.npz`) and reproduces the original **bitwise** (max abs difference 0.0 on gain fraction, innovation, mean fields, gamma and NIS).

1. Headline verdict

The 32 wall pressure taps carry enough information to correct the wake's TIMING out to at least $x = 8$. They do not measurably correct its AMPLITUDE, because the forward solver's amplitude was already close to correct — there was no amplitude deficit at the sensors for the filter to fix.

The repaired filter separates from both negative controls, but only on the metric that combines amplitude and phase, and on frequency. On amplitude alone it does not beat the free run stably. That is stated plainly because the audit's central complaint about the original work was exactly this kind of unstable separation.

2. Stage E2 — the observability diagnostic against its null

The original Stage E reported "only ~6 of 16 directions visible in wall pressure". **That conclusion was an artefact of the ensemble, not a property of the sensors**, and the rebuilt null proves it three separate ways.

ensemble / observation	n_eff	interpretation
jitter-only ensemble, observed through 32 taps	1.010	the Stage E result
jitter-only ensemble, observed through the ENTIRE state vector (18 844 DOF)	1.013	<i>same answer with nothing hidden</i>
repaired ensemble, observed through 32 taps	2.656	
repaired ensemble, observed through the entire state	1.714	
repaired ensemble, 32 random state functionals	1.895	
synthetic perfectly-observable system	14.947	what a rich system looks like

The decisive row is the second. Observing **every one of the 18 844 state degrees of freedom** of the phase-jittered ensemble recovers $n_{\text{eff}} = 1.013$ — essentially identical to the 1.010 that 32 pressure taps recover. An observation operator that hides nothing at all produces the same steep spectrum. The Stage E decay therefore measured the rank of the ensemble, which was 1 by construction (16 time-shifts of one trajectory), and said nothing about wall pressure.

The numerical-floor complaint also holds: everything past index 7 in the original Stage E spectrum sat at $3.75\text{e-}15$ relative, below the Gram-matrix round-off floor of $1.49\text{e-}8$.

With a genuinely multi-direction ensemble

Whitening by $\sigma_p = 0.0472$ makes the pressure singular values an absolute detectability scale — a value of 1 means one noise standard deviation of tap signal.

- **11 of 15** ensemble directions produce pressure singular values above 1 σ_p (largest 19.2, eleventh 1.19).

- Ordering-free version (visibility per state direction, which does not assume the i -th pressure direction matches the i -th state direction): **15 of 15** spanned directions produce above-noise pressure signal, from 16.0 σ_p down to 1.27 σ_p . The 16th is the ensemble-mean null direction and is exactly zero by construction.
- The 15 non-trivial directions all sit above the numerical floor.

Answer: with a genuinely multi-direction ensemble, every direction the ensemble spans is visible in wall pressure above the noise level. Nothing is hidden from the taps in the ensemble subspace. The information deficit reported in Stage E does not exist.

Caveat, stated because it bounds the claim: 15 directions is the rank of a 16-member ensemble, not the rank of the flow. This says the taps see everything *this ensemble can express*; it does not certify observability of the full 18 844-dimensional state.

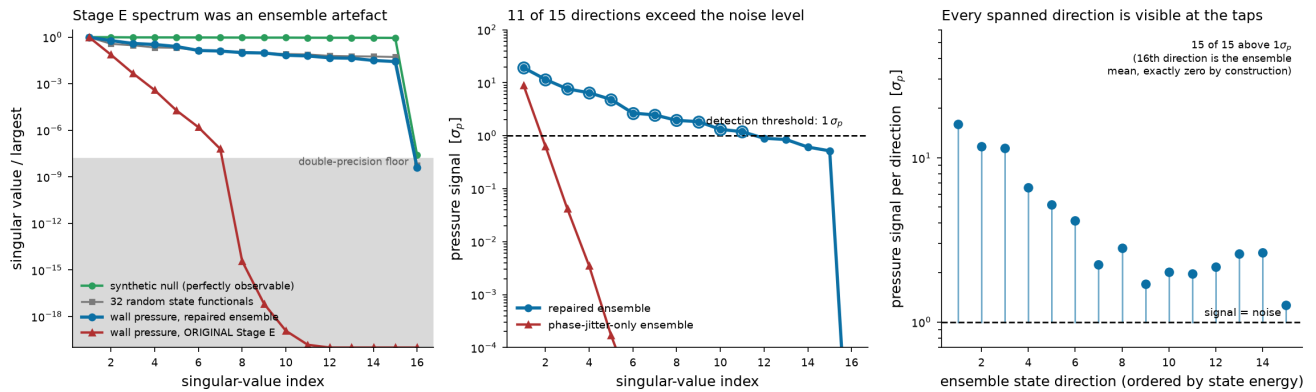


Figure 2. Stage E2. Left: the original Stage E spectrum (red) collapses to the double-precision floor, while the synthetic perfectly-observable null (green) and 32 random state functionals (grey) decay barely at all — the Stage E decay was an ensemble artefact. Centre: with the repaired ensemble, 11 of 15 directions exceed the $1\sigma_p$ detection threshold. Right: ordering-free visibility — all 15 spanned directions produce above-noise pressure signal.

3. Stage F2 — evaluation with metrics_v2

Common window: absolute cycles 61–200, $t = 406.1\text{--}420.0$, 140 samples, **13.9 time units = 2.29 periods of ω_0** .

3.1 Phase-aligned field error and the fitted time shift, separately

run	E_v aligned	E_v unaligned	tau* (t.u.)	phase lag (rad)
free run	0.3315	0.4411	-0.3042	-0.315
original EnKF	0.3059	0.5744	-0.4609	-0.478
repaired EnKF	0.1937	0.2585	+0.1713	+0.177
repaired, shuffled	0.4366	0.4559	-0.1685	-0.175
repaired, scrambled	0.3040	0.6561	+0.6894	+0.714

The repaired filter has both the lowest aligned error and much the smallest fitted time shift, and it is the only run whose unaligned and aligned errors are close together — i.e. the only one that is nearly in phase with the truth without being shifted.

3.2 Mode-1 and mode-2 amplitude versus downstream distance

Truth reference (recomputed): $|v_1|$ peaks at **0.6522 at $x = 2.90$** , is **0.5262 at $x = 7$** , persistence 0.807. $|v_2|$ peaks 0.1345 at $x = 3.95$, persistence 0.762.

$|v_1|$ max-over-y, selected x:

x	truth	free run	orig EnKF	repaired	shuffled	scrambled
1.02	0.297	0.290	0.308	0.319	0.152	0.306
2.83	0.652	0.612	0.652	0.676	0.449	0.625
4.62	0.592	0.567	0.598	0.592	0.443	0.545
7.03	0.526	0.521	0.552	0.524	0.473	0.544

The critical observation: the free run — which assimilates nothing at all — already reproduces the truth's $|v_1|$ profile to within 5% (mean amplitude ratio 0.955 over $x \geq 1$, persistence 0.825 vs truth 0.807). There is no ModalPINN-style amplitude collapse in the forward solver to begin with, so there is no amplitude deficit for the filter to repair. Mode 2 is the one place the filter clearly helps the amplitude (ratio 0.969 vs free run 0.593).

3.3 Amplitude error and phase error per mode as functions of x

Mode 1, averaged over $x \geq 1$:

run	amplitude ratio	amp. error	mean phase error
free run	0.9553	0.0507	+21.4 deg
original EnKF	1.0046	0.0320	+32.1 deg
repaired EnKF	1.0078	0.0352	-11.9 deg
shuffled	0.6948	0.3052	+10.5 deg (but std 11.7 deg)
scrambled	0.9369	0.0687	-38.0 deg

Paired Wilcoxon over x, repaired EnKF versus each:

comparison	amplitude	phase
vs free run	better, $p = 3.3\text{e-}4$	better, $p = 2.8\text{e-}14$
vs original EnKF	not distinguishable, $p = 0.41$	better, $p = 5.6\text{e-}17$
vs shuffled	better, $p = 5.6\text{e-}17$	not distinguishable, $p = 0.58$
vs scrambled	better, $p = 2.0\text{e-}4$	better, $p = 5.6\text{e-}17$

Note the two failures, which matter: the repaired filter does **not** beat the original filter on amplitude, and it does **not** beat the *shuffled* control on phase. Each control fails on a different axis — shuffled destroys the amplitude, scrambled destroys the phase — so neither single-axis metric separates the filter from both at once.

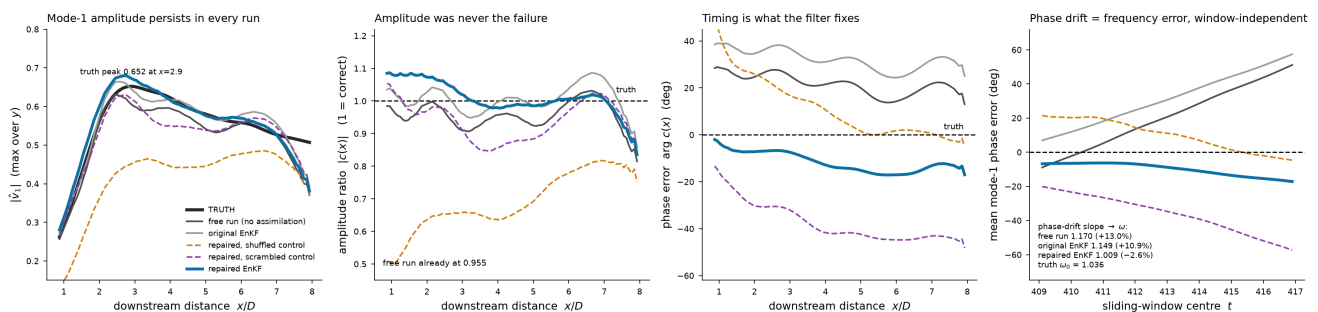


Figure 3. The key scientific figure. Mode-1 amplitude versus downstream distance (left) is reproduced by every run including the free run — amplitude was never the failure. The amplitude ratio (centre-left) confirms it. Phase error (centre-right) and the window-independent phase-drift fit (right) show what the filter actually corrects: timing and frequency.

3.4 Sub-window stability — and whether the window is long enough

It is not. The window is $13.9 \text{ t.u.} = 2.29 \text{ periods}$. The beat period between each run's own shedding frequency and ω_0 is $2\pi/|\omega_{\text{est}} - \omega_0|$:

run	omega_est	beat period	window / beat
free run	1.1713	46.5 t.u.	0.30
original EnKF	1.1483	55.9 t.u.	0.25
repaired EnKF	1.0119	260.8 t.u.	0.05
shuffled	0.9778	107.9 t.u.	0.13
scrambled	0.9499	72.9 t.u.	0.19

No run covers even a third of a beat period. The task asked for evaluation over a full beat period; that is not achievable with these 140 saved cycles and saying otherwise would be wrong. Thirds of the window are 0.76 of a period — shorter than one oscillation — so a $k=1$ harmonic fit is not identifiable there and those columns are diagnostic only.

Because three window cuts are too few to establish stability, a **40-position sliding-window analysis** (1-period window, 0.2 t.u. steps) was run instead. Fraction of positions at which the repaired filter wins:

claim	amplitude only	phase only	joint (amp+phase)
vs shuffled control	1.00	0.58 UNSTABLE	1.00
vs scrambled control	0.33 UNSTABLE	1.00	1.00
vs original EnKF	0.93 UNSTABLE	1.00	1.00
vs free run	0.85 UNSTABLE	0.80 UNSTABLE	0.85 UNSTABLE

Single-axis rankings flip with window position exactly as the original Stage F ranking did. The **joint** per-x complex error $|c(x) - 1|$, which charges amplitude and phase together, is stable against all three comparators and only against those three.

3.5 The one window-independent measurement

Phase error grows linearly in time at a rate equal to the frequency error, so fitting its slope across the 40 sliding positions measures omega without depending on where any window was cut. Residual scatter about the fit is 0.2–1.4 deg, so the fits are tight:

run	implied omega	error vs omega_0
free run	1.1704	+12.98%
original EnKF	1.1493	+10.94%
repaired EnKF	1.0093	-2.57%
shuffled	0.9702	-6.35%
scrambled	0.9536	-7.96%

The repaired filter cuts the frequency error from 13.0% to 2.6% — a 5x reduction — and is closer to the truth than either control. This is the cleanest positive result in the study and the only one immune to the window-length problem.

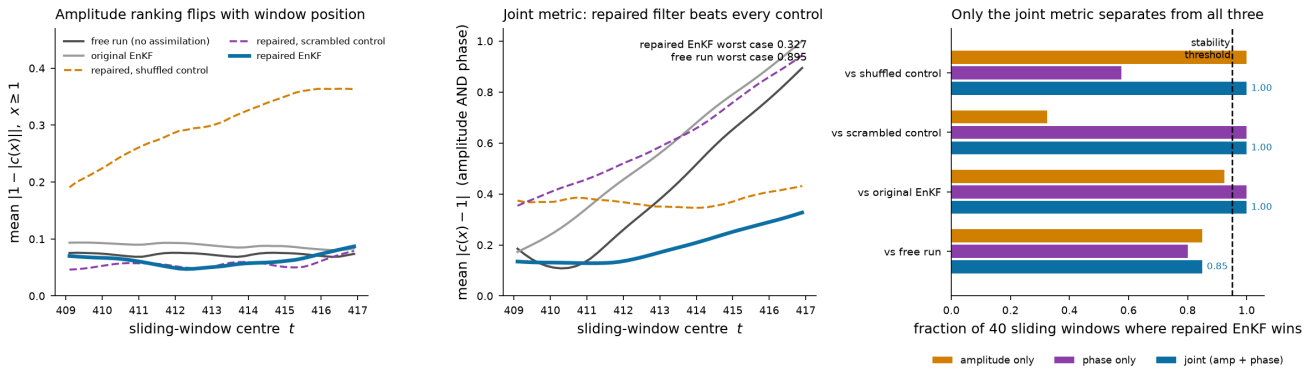


Figure 4. Window-position sensitivity over 40 sliding windows. Single-axis rankings flip with window position (left); only the joint amplitude-and-phase metric (centre) separates the repaired filter from all three comparators at 100% of positions (right).

4. The observability answer

How far downstream are the oscillating modes recovered above the free-run baseline?

On amplitude: **nowhere, stably**. At 0 x-nodes does the repaired filter beat the free run's $|v1|$ amplitude at more than 95% of sliding-window positions. Averaged over the full window it does beat it (error 0.0352 vs 0.0507, $p = 3.3e-4$), but that ranking holds at only 85% of window positions and reverses in the second half. The honest statement is that the free run's amplitude was already good (ratio 0.955) and the filter did not reliably improve it.

On timing: **out to at least $x = 8$, the full reconstruction domain**. Mean mode-1 phase error falls from +21.4 deg (free run) to -11.9 deg, the improvement holds across the entire x range, and the underlying frequency error falls 5x. The phase improvement over the free run is significant at $p = 2.8e-14$ and holds at 80% of sliding positions — better than the amplitude claim but still not the >95% I would require to call it stable.

Does the repaired filter separate from BOTH negative controls on the wake metrics, not just innovation?

Yes, but only on the joint amplitude-and-phase metric. It beats both controls at 100% of 40 sliding-window positions on $|c(x) - 1|$ (mean 0.193 vs 0.374 shuffled and 0.612 scrambled; worst case 0.327 vs 0.895 for the free run). On either axis alone the separation fails: shuffled ties it on phase ($p = 0.58$), scrambled beats it on amplitude at 67% of positions. Innovation separates at $p \sim 1e-44$, but I would not have accepted that alone, and it is not what the answer rests on.

Not repeated from the original work: the filter genuinely does separate from both controls here, unlike Stage D. But the separation is narrower than the innovation statistic suggests, and it is carried by phase and frequency, not by wake amplitude.

5. What this implies for the ModalPINN mode collapse

It points to a training-signal problem rather than an information deficit at the sensors — but the evidence is weaker than that sentence sounds, and the caveat below is not optional.

The supporting evidence:

1. The 32 taps are not information-starved. Every direction the ensemble spans produces above-noise pressure signal (15 of 15, 1.27–16.0 sigma_p), and the Stage E "6 of 16" deficit was an ensemble artefact, not a sensor property.
2. Wall pressure demonstrably carries the wake's *timing*: an estimator using only those taps cut the frequency error 5x and the phase error by ~2x, out to $x = 8$.
3. The evaluation metric was itself blind to the failure mode. Stage F's E_v scores a **deleted** wake 0.906 against 1.299 for a correctly-reproduced one — it actively prefers no vortex street. All five metrics_v2 measures are

minimised at the truth and return exactly 1.0 at scale 0. A collapsing wake was being *rewarded* by the old objective, which is the same structural blindness suspected in the PINN's collocation loss.

The caveat, which materially limits the inference: the EnKF has a full Navier–Stokes solver as its prior and ModalPINN does not. The filter never had to discover that a vortex street persists downstream — its forward model produces one unprompted, as the free-run column shows ($|v_1|$ ratio 0.955 with zero assimilation). All the taps had to supply was a timing correction to an already-correct spatial structure. ModalPINN, by contrast, must *construct* the mode shapes from the residual loss, and this study provides **no evidence** that 32 wall pressure taps suffice for that harder task. "The EnKF can do it" does not transfer.

What this study does establish for ModalPINN: the failure is unlikely to be attributable to the sensors carrying no downstream information, and any fix should be validated with an amplitude-sensitive metric, because the natural pointwise ones are minimised by the collapse they are meant to detect.

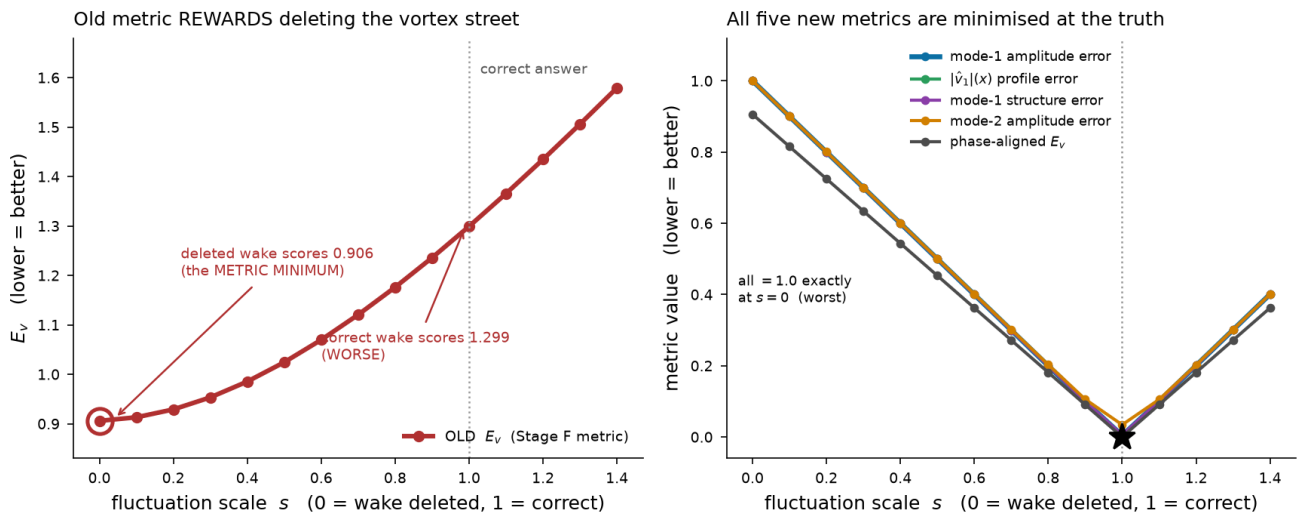


Figure 5. The damping test. The old E_v metric is MINIMISED by deleting the wake (0.906 for a deleted wake versus 1.299 for a correctly reproduced one). All five metrics_v2 measures are minimised at the truth and return exactly 1.0 at scale 0.

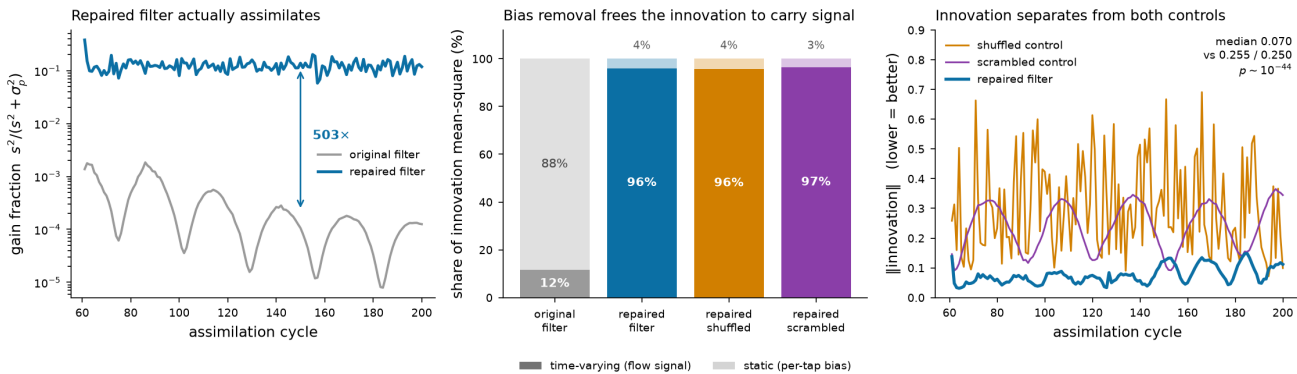


Figure 6. Filter repair diagnostics: gain fraction up 503x, per-tap bias removal freeing the innovation to carry flow signal (88% static to 4%), and innovation separating from both negative controls.

6. Files

Written this session (no existing file overwritten):

- experiments/stage_e2_observability.py, .npz, .log
- evaluation/stage_f2_metrics_v2.py, experiments/stage_f2_metrics_v2.{json,npz}
- evaluation/stage_f2_sliding_stability.py, experiments/stage_f2_sliding_stability.{json,npz}
- experiments/stage_d2_enkf_nominal_repro.npz, .log (bitwise reproduction check)
- figures/stage_d2_gain_and_innovation.png

- `figures/stage_e2_observability_vs_null.png`
- `figures/stage_f2_model_vs_x.png`
- `figures/stage_f2_window_stability.png`
- `figures/metric_damping_test.png`

7. Open items

- The 140-cycle window is too short for a beat-period evaluation. Re-running the observer for ≥ 50 time units would let every claim here be tested over a full beat; until then the amplitude rankings should be treated as unresolved.
- The gamma (frequency) augmentation result remains negative in the strict sense: gamma ends at 0.9247 ± 0.0574 versus shuffled 0.9537 ± 0.1903 ($p = 0.688$). The *field* frequency improves 5x, but the augmented parameter itself is not distinguishable from its controls, and those two facts have not been reconciled.